

Locomotion Capabilities of a Novel Reconfigurable Robot with 3 DOF Active Joints for Rugged Terrain

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Abstract - Reconfigurable robots consist of many modules which are able to change the way they are connected. As a result, the robots have the capability of adopting different configurations to match various tasks and suit complex environments. This paper presents a novel field robot named JL-I which consists of three uniform modules. With the docking mechanisms, the modules can connect or disconnect flexibly and automatically. Furthermore the active joints formed by serial and parallel mechanisms endow the robot with the ability of changing shape in three dimensions. Consequently useful locomotion capabilities, such as crossing high vertical obstacles, getting self-recovery when the robot is upside-down are achieved. After describing the structural principle of the robot, the related kinematics analysis follows. In the end, the successful on-site tests confirm the principles described above and the robot's ability.

Index Terms - Reconfigurable robot, Docking mechanism, Spherical joint, Locomotion capabilities

I. INTRODUCTION

The last decade has seen an increasing interest in developing and employing mobile robots for industrial inspection, conducting surveillance, urban search and rescue, military reconnaissance and civil exploration not only indoors but also outdoors. The reconfiguration is very essential in such tasks which are difficult for a fixed-shape robot and during which robots have to confront unstructured environments [1]. The basic requirement for this kind of robotic systems moving on rough terrains is extraordinary motion capabilities including three aspects:

- 1) The ability of moving over all working areas: The robot should have a flexible mobility in the rugged terrain in order to get to every point in the working space.
- 2) The ability of dealing with obstacles: In order to finish a task in an unstructured environment, the ability to cross high obstacles and span large gaps is definitely necessary.
- 3) The ability of crossing narrow openings: Sometimes the working environment is very complicated, which includes not only high steps and deep ditches but also narrow fences and floors cluttered with debris. The robot has to handle uncertainties and cross narrow openings by reconfiguring its structure.

A lot of successful research has been carried out in this field. The common kinematics modes include multiple legs, sliding frames, wheeled and chain-track vehicles. Mark Yim [2] proposed the reconfigurable robot PolyBot for urban

searching and rescuing. It is able to optimize the way its parts are connected to fit the specific task. PolyBot adopts its shape to become a rolling type for passing over flat terrain, an earthworm type to move in a narrow space and a spider type to stride over uncertain hilly terrain. Another wheeled serpentine robot from Takayama and Hirose consists of three segments. Each segment is driven by a pair of tracks, but all tracks are powered simultaneously by a single motor located in the centre segment [3]. The special ability of adapting to irregular terrain is passive and provided by springs.

The application of the modular reconfiguration concept to field robots enriches their configurations and improves their adaptability to different environments [4]-[9]. However, the known robots usually have few configurations due to relatively simple docking and pose-adjusting mechanisms. For example, a reconfigurable mobile robot designed by M. Park is not able to change its configuration actively at all [10]. A modular tracked robot proposed by T. Wang [11] has only one DOF in its pose-adjusting mechanism. The robot from Université Libre de Bruxelles [12] has a one-DOF pose-adjusting mechanism and one coupler to change the configuration between the neighbouring modules as well.

Since 2000 our group has been focusing on the design and development of a modular reconfigurable mobile robot. A smart structure with two linked-track vehicles was proposed by W. Wang [13]. The structure can be reconstructed so that the robot can move between surfaces with an angle of 0 - 90 degrees. However, the construction stiffness is very low so that there is a small distortion while moving.

This following project is intended to develop an automatic field robot for unstructured environments to meet the requirements of high flexibility, robustness and low cost. It is necessary to increase the locomotion capabilities of the robot. Based on this consideration, a novel modular reconfigurable mobile robot named JL-I was developed, which has three uniform modules. After giving an overview of the robot, the discussion focuses on the locomotion capabilities. The related kinematics analysis of the mechanical realization is discussed thoroughly as well. In the end, a series of successful on-site tests is presented to confirm the principles described above and the robot's ability.

II. THE OVERVIEW OF THE RECONFIGURABLE JL-I

The system consists of three connected, identical modules for crossing grooves, steps, obstacles and moving in a

complex environment. The mechanical structure is flexible due to its uniform modules and special connecting joints (Fig. 1).

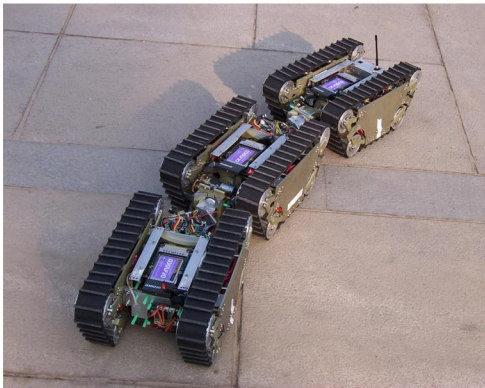


Fig. 1. A photo of JL-I.



Fig. 2. Performing distributed activities.

Actually each module is an entire robotic system that can perform distributed activities (Fig. 2). Three DOF active spherical joints between the modules and the docking mechanism enable the adjacent robots to adopt optimized configurations to negotiate difficult terrains or to split into several small units to perform tasks simultaneously. In contrast to the results of conventional former research, this robot includes the following innovative aspects:

1. Smooth movement: It is known that a common mobile robot will lose its moving ability if it is not able to keep its balance. However, the robot JL-I with many active spherical joints can smooth the undulating motion of the gravity centre of the whole robot.

2. High adaptability: Identical modules in the JL-I robot have a large variety of configurations owing to the pose-adjusting joints and the docking mechanisms. Therefore, the robot can adopt many shapes which make the JL-I system able to carry out various different tasks and move in diverse environments.

3. Self recovering ability: Furthermore, JL-I has the ability to implement self-recovery. By disconnecting the malfunctioning module, the system can repair itself if one module does not work normally. In this way, working safety and efficiency are improved a lot when the robot moves in a complex environment.

The single module is about 35 centimetres long, 25 centimetres wide and 15 centimetres high. Fig. 3 shows the mechanical structure of the single module which has two

powered tracks, a serial mechanism, a parallel mechanism, and a docking mechanism. Two DC motors drive the tracks providing skid-steering ability in order to realize the flexible omnidirectional movement. The docking mechanism consists of two parts: a cone-shaped connector at the front and a matching coupler at the back of the module. It enables any two adjacent modules to link and form a train configuration.

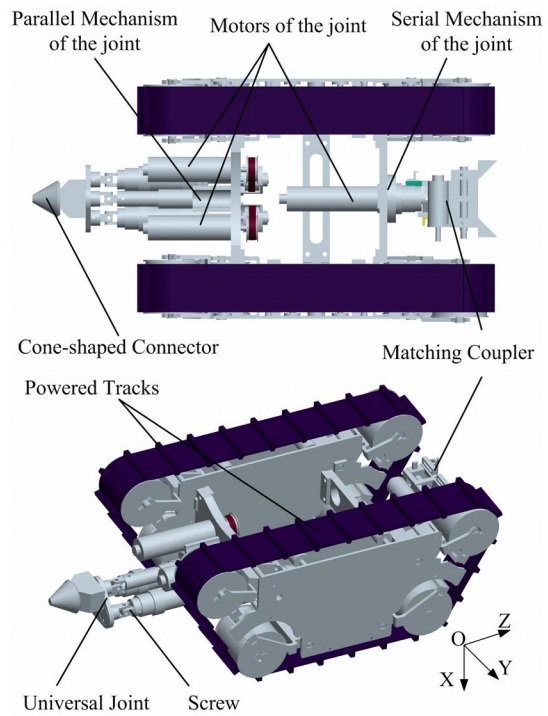


Fig. 3. The artistic impression of the module.

The robot features the serial and parallel mechanisms which form a three-DOF active spherical joint. There are two reasons for using the serial and parallel mechanisms for our mobile robot. Firstly, the JL-I robot can be made lightweight and dexterous while allowing for a larger payload. Secondly, the advantages of the high rigidity of a parallel mechanism and the extended workspace of a serial mechanism can be combined, thus improving the flexibility of the robotic system. By controlling the active joints and the docking mechanisms, the robot can change its shape in three dimensions. The docking mechanism is capable of self-aligning within certain lateral and directional offsets. The serial mechanism can rotate 360° around the Z axis and the parallel mechanism can pitch around the Y axis and yaw around the X axis.

To ensure its ability of performing tasks individually, there is enough space in each module for sensors, the onboard controller, and batteries. Multiple sensing and control systems are incorporated to handle uncertainties in the complex environment. The Software is dexterous enough to identify the various geometries and intelligent enough to autonomously reconstruct the environment. In order to meet the requirements of a light weight and a dexterous movement mechanism, considerable stress is laid on weight reduction as well as on

construction stiffness. Most of the mechanical parts are designed specifically and mainly manufactured in aluminium. It weighs approximately 7 kg including the batteries.

III. LOCOMOTION CAPABILITIES

The proposed mobile system should have various moving modes. By combining such locomotion capabilities, JL-I will move in almost all kinds of rough environments. Because of the ambiguity in the use of the schemes below, we introduce the following definitions for the remainder of this paper. From the top view of Fig. 1, the robot will be illustrated as in Fig. 4a. From the lateral view, the robot will be illustrated as in Fig. 4b.

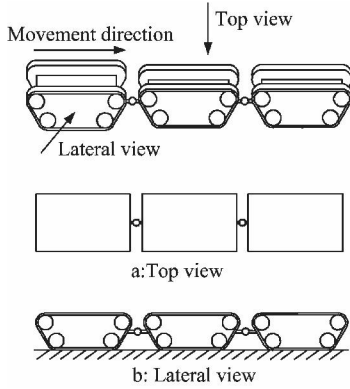


Fig. 4 The illustration of movements

A. Terrain Adaptability

The principle of terrain adaptability is shown in Fig. 5. The robot can change its posture by pitching around the Y axis, yawing around the X axis and rotating around the Z axis. That means it is possible for the robot to obtain high mobility over rugged terrain.

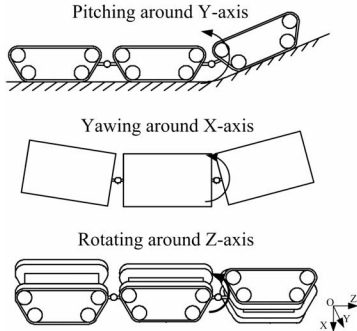


Fig. 5. Adapting to terrain by pitching, yawing and rotating.

B. Crossing a Step

Fig.6 shows the process of crossing a step from a view in the Y direction. Here the step is almost twice as high as the robot.

- The robot is in its home state, and the sensor detects the step in the movement direction.
- The first module is pitching up around the Y axis while the robot is moving forward.
- The approaching movement does not stop until the first module touches the step.

(d) The first module pitches down to attach to the top of the step.

(e) The robot is moving forward continuously.

(f) The robot is moving until the first two modules are attached to the step.

(g) The last module is pitching up around the Y axis while the robot is moving forward.

(h) The robot is now in its home state again, and the process of crossing the step is over.

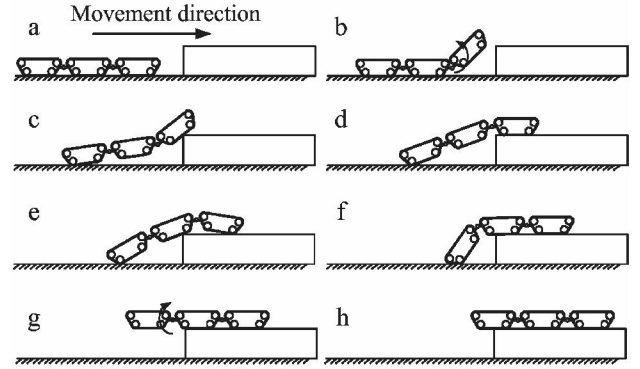


Fig. 6. The sequence of crossing a step.

C. 90° Self-recovery

It is possible for the robot to implement a 90° recovering movement by adopting the proper configuration sequence as shown in Fig. 7.

(a) The robot is lying on its side

(b) The first module and the last module are yawing up around the X axes of the active joints.

(c) Then the first module and the last module rotate 90° around the Z axes.

(d) After that, they are pitching down around the Y axes of the active joints until they attach to the ground in order to raise the middle module up.

(e) The middle module rotates around the Z axis until it is parallel to the ground.

(f) In the end, the module is pitching down around the Y axes of the active joints until all three modules attach to the ground together. The robot is now in its home state again, and the process of 90° Self-recovery is over.

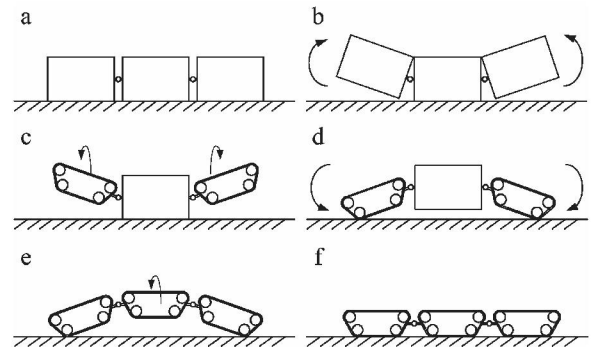


Fig. 7. 90° recovering movement.

D. 180° Self-recovery

It is also possible for the robot to tip over and realize the 180° recovery movement as shown in Fig. 8.

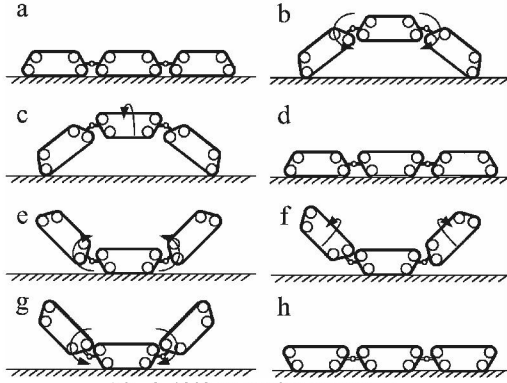


Fig. 8. 180° recovering movement.

- The robot is in its home state.
- The first and the last modules are pitching down around the Y axes of the active joints until the middle module is in the air.
- The middle module rotates 180° according to the Z axis.
- The first module and the last module are pitching down around the Y axes of the active joints until the middle module attaches to the ground.
- The first module and the last module are pitching up around the Y axes of the active joints again.
- The first module and the last module are rotating 180° around the Z axes of the active joints.
- Then the first module and the last module are pitching down around the Y axes of the active joints again until all three modules attach to the ground.
- The process of 180° Self-recovery is over.

E. Crossing a Narrow Fence

As shown in Fig. 9, the train configuration robot is able to cross a fence narrower than the width of its modules.

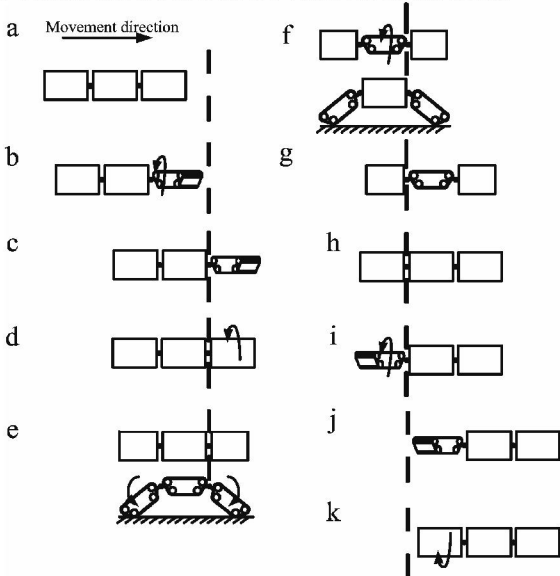


Fig. 9. The sequence of crossing a narrow fence.

(a) The robot is in its home state, and the sensor detects the fence in the moving direction.

(b) The robot stops before the fence, and then the first module pitches up around the Y axis and then rotates 90° according to the Z axis.

(c) The crossing movement does not stop until the first module passes through the fence.

(d) The first module rotates and pitches to get back into the home state, and then the three modules attach to the ground together again.

The following steps (e) to (k) of the second and third modules are similar to those of the first one. The process will be achieved until the robot crosses the fence entirely. In order to show the principle clearly, the lateral views of steps (e) and (f) are also given.

IV. KINEMATICS ANALYSIS OF THE ACTIVE SPHERICAL JOINT

As described above, the robot's reconfiguring abilities are achieved by the motion of the 3 DOF active spherical joints. Two of the DOF achieved by the parallel mechanism are yawing and pitching around the joint's X and Y axes respectively. The third rotation DOF around the joint's Z axis is achieved by the serial mechanism.

To demonstrate the reconfiguring possibility, the kinematics analysis of two connected modules should be studied first. Fig. 10 shows the kinematics model of the joint between two modules. Where $OXYZ$ is the world coordinate fixed at the plane QEF which represents the front unmovable module during the reconfiguration. The origin is located at the universal joint O , the Z-axis coincides with the axis of the serial mechanism and the X-axis points to the middle point of line AB . Another reference coordinate $O'X'Y'Z'$ is fixed at triangular prism $OABPCD$ which represents the back moveable module. The $O'X'Y'Z'$ is coincident with the $OXYZ$ when the spherical joint is in its home state. Equations (1) and (2) are satisfied due to the mechanical constraints. QF is perpendicular and equal to QE .

$$\begin{aligned} QEF // OAB // PCD \\ QEF = OAB = PCD \end{aligned} \quad (1) \quad (2)$$

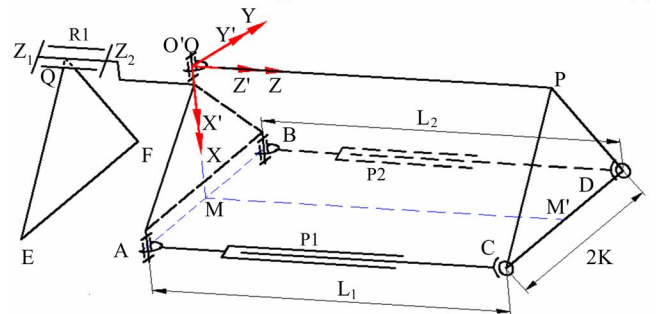


Fig. 10. The kinematics model of the active spherical joint.

The required orientation for the reference frame $O'X'Y'Z'$ on the back module is achieved by a rotation of θ_z , a pitching angle θ_y and a yawing angle θ_x according to the relative axes. From the mechanical point of view, actually the pitching and yawing motions are realized by the outstretching and returning

movement of the L_1 , L_2 of the parallel mechanism, and the rotation of θ_z is actuated by the serial mechanism. The freedom of the reconfiguring movement is three and can be described with the generalized coordinate θ (3). The joint variants of the movement are named q , described as (4).

$$\theta = [\theta_x, \theta_y, \theta_z]^T \quad (3)$$

$$q = [L_1, L_2, \theta_z]^T \quad (4)$$

The purpose of the kinematics analysis is to deduce the relationship between q and θ . In Fig. 10, the points A , B , C , D are described as (5) in the $OXYZ$ coordinate.

$$A = \begin{bmatrix} K \\ -K \\ 0 \end{bmatrix} \quad B = \begin{bmatrix} K \\ K \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} K \\ -K \\ L \end{bmatrix} \quad D = \begin{bmatrix} K \\ K \\ L \end{bmatrix} \quad (5)$$

The homogeneous transformation matrix $[T]$ from the world coordinate $OXYZ$ to the coordinate $O'X'Y'Z'$ is described as (6).

$$T = Rot(Y)Rot(X)Rot(Z) = \begin{bmatrix} c\theta_y & 0 & s\theta_y \\ 0 & 1 & 0 \\ -s\theta_y & 0 & c\theta_y \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta_x & -s\theta_x \\ 0 & s\theta_x & c\theta_x \end{bmatrix} \begin{bmatrix} c\theta_z & -s\theta_z & 0 \\ s\theta_z & c\theta_z & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (6)$$

After the reconfiguring movement, A , B , C , and D are changed to new positions described as A_1 , B_1 , C_1 , and D_1 . The Cartesian coordinates of the new points can be expressed as (7) and (8).

$$[A_1 \ B_1] = Rot(Z)[A \ B] \quad (7)$$

$$[C_1 \ D_1] = T[C \ D] \quad (8)$$

There are some constraints to the mechanical structure, as shown in (9) and (10). The lengths of the link L_1 and L_2 are equal to the distance between C_1A_1 and D_1B_1 respectively.

$$L_1 = C_1 - A_1 \quad (9)$$

$$L_2 = D_1 - B_1 \quad (10)$$

All these results are inserted into (9) and (10); resulting in equation (11).

$$\begin{aligned} [L_1 \ L_2] &= [C_1 \ D_1] - [A_1 \ B_1] = \\ &\begin{bmatrix} K(c\theta_y c\theta_z + s\theta_x s\theta_y s\theta_z) - K(-s\theta_z c\theta_y + s\theta_x s\theta_y c\theta_z) + Lc\theta_x s\theta_y - Kc\theta_z - Ks\theta_z \\ K(c\theta_x s\theta_z) - K(c\theta_x c\theta_z) - Ls\theta_x - Ks\theta_z + Kc\theta_z \\ K(-s\theta_y c\theta_z + s\theta_x c\theta_y s\theta_z) - K(s\theta_y s\theta_z + s\theta_x c\theta_y c\theta_z) + Lc\theta_x c\theta_y \\ K(c\theta_y c\theta_z + s\theta_x s\theta_y s\theta_z) + K(-s\theta_z c\theta_y + s\theta_x s\theta_y c\theta_z) + Lc\theta_x s\theta_y - Kc\theta_z + Ks\theta_z \\ K(c\theta_x s\theta_z) + K(c\theta_x c\theta_z) - Ls\theta_x - Ks\theta_z - Kc\theta_z \\ K(-s\theta_y c\theta_z + s\theta_x c\theta_y s\theta_z) + K(s\theta_y s\theta_z + s\theta_x c\theta_y c\theta_z) + Lc\theta_x c\theta_y \end{bmatrix} \end{aligned} \quad (11)$$

The kinematics expression also results from this, described as (12) (13).

$$\begin{aligned} L_1^2 &= (K(c\theta_y c\theta_z + s\theta_x s\theta_y s\theta_z) - \\ &\quad K(-s\theta_z c\theta_y + s\theta_x s\theta_y c\theta_z) + \\ &\quad Lc\theta_x s\theta_y - Kc\theta_z - Ks\theta_z)^2 + \\ &\quad (K(c\theta_x s\theta_z) - K(c\theta_x c\theta_z) - Ls\theta_x - Ks\theta_z + Kc\theta_z)^2 + \\ &\quad (K(-s\theta_y c\theta_z + s\theta_x c\theta_y s\theta_z) - \\ &\quad K(s\theta_y s\theta_z + s\theta_x c\theta_y c\theta_z) + Lc\theta_x c\theta_y)^2 \\ &= T_{L_1} \end{aligned} \quad (12)$$

$$\begin{aligned} L_2^2 &= (K(c\theta_y c\theta_z + s\theta_x s\theta_y s\theta_z) + \\ &\quad K(-s\theta_z c\theta_y + s\theta_x s\theta_y c\theta_z) + \\ &\quad Lc\theta_x s\theta_y - Kc\theta_z + Ks\theta_z)^2 + \end{aligned}$$

$$\begin{aligned} &(K(c\theta_x s\theta_z) + K(c\theta_x c\theta_z) - Ls\theta_x - Ks\theta_z - Kc\theta_z)^2 \\ &+ (K(-s\theta_y c\theta_z + s\theta_x c\theta_y s\theta_z) + \\ &\quad K(s\theta_y s\theta_z + s\theta_x c\theta_y c\theta_z) + Lc\theta_x c\theta_y)^2 \\ &= T_{L_2} \end{aligned} \quad (13)$$

Named $T_{L1}=L_1^{1/2}$, $T_{L2}=L_2^{1/2}$, then the relation between q and θ can be concluded as (14).

$$q = [T_{L1}^{1/2}, T_{L2}^{1/2}, \theta_z]^T \quad (14)$$

The relationship of the world coordinate and the reference joint coordinate can be concluded. Furthermore the movements can be anticipated according to the joints' driving outputs.

V. ON-SITE EXPERIMENTS

Relevant successful on-site experiments with the mobile robot were carried out recently, confirming the principles described above and the robot's ability. Fig. 11 shows the docking process of the connection mechanism whose most distinctive features are its ability of self aligning and its great driving force. With the help of the powered tracks, the cone-shaped connector and the matching coupler can match well within $\pm 30\text{mm}$ lateral offsets and $\pm 45^\circ$ directional offsets.

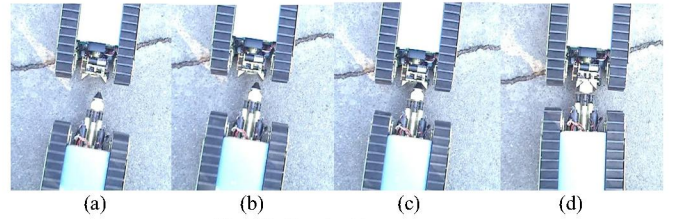


Fig. 11. The docking process.

Compared with many configurable mobile robots, the JL-I improves its flexibility and adaptability by using novel active spherical joints between modules. The following figures show the typical motion functionalities one by one, whose principles are discussed above.

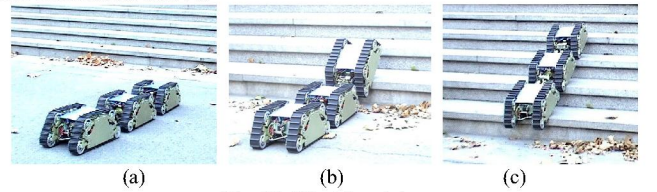


Fig. 12. Climbing stairs.

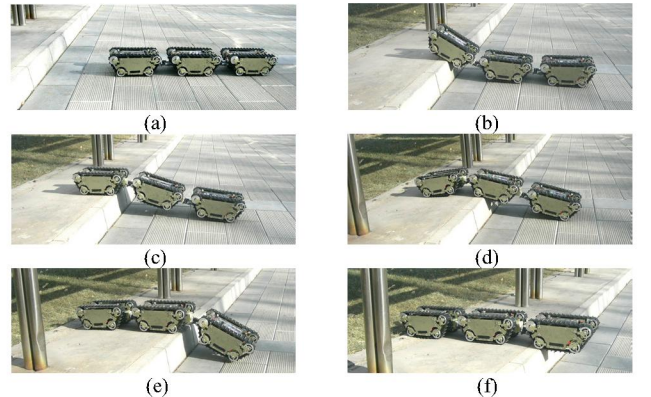


Fig. 13. Snapshots of crossing a step

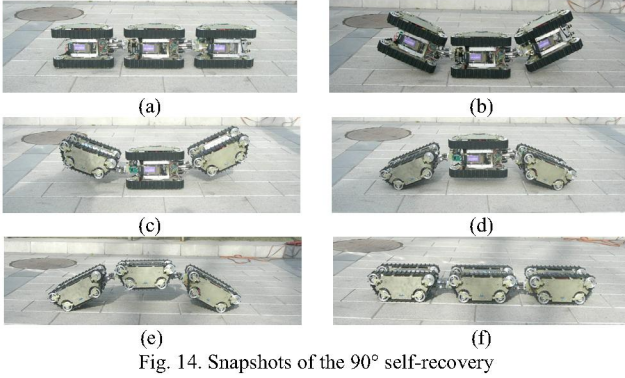


Fig. 14. Snapshots of the 90° self-recovery

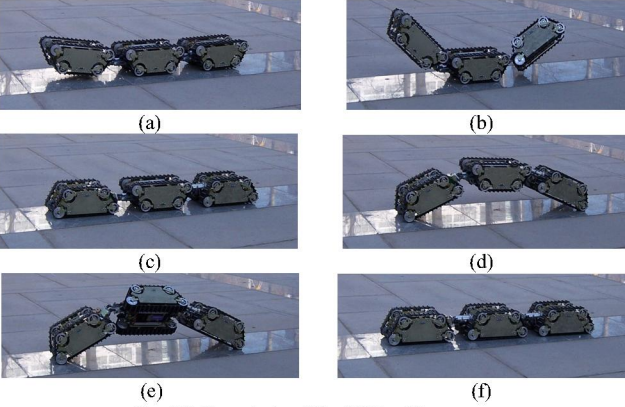


Fig. 15. Snapshots of the 180° self-recovery

The experimental results show that the 3 DOF active joints with serial and parallel mechanisms have the ability to achieve all the desired configurations. The performance specifications of JL-I are given in TABLE I.

TABLE I
PERFORMANCE SPECIFICATIONS

Posture adjustment angle around X-axis:	$\pm 45^\circ$
Posture adjustment angle around Y-axis:	$\pm 45^\circ$
Posture adjustment angle around Z-axis:	$0 \sim 360^\circ$
Maximum lateral docking offset:	± 30 mm
Maximum directional docking offset:	$\pm 45^\circ$
Maximum height of steps:	280 mm
Maximum length of ditches:	500 mm
Minimum width of the fence:	200 mm
Maximum slope angle:	40°
Self-recovering ability:	$0 \sim 180^\circ$
Maximum climbing velocity:	180 mm/s
Maximum unchangeable working time:	4 hours

VI. CONCLUSIONS

The modular reconfiguration robot has the ability of changing its configuration which makes it more suitable for complex environments. In contrast to conventional theoretical research, the project introduced in this paper successfully completes the following innovative work:

1. It proposes a robot named JL-I which is based on a modular reconfiguration concept. The advantages and the characteristics of the mechanism are analysed. The robot

features a docking mechanism with which the modules can connect or disconnect flexibly. The active spherical joints formed by serial and parallel mechanisms endow the robot with the ability of changing shapes in three dimensions.

2. A kinematics model of reconfiguration between two modules is given. The relationship of the world coordinate and the reference joint coordinate is concluded. Furthermore, the movements can be anticipated according to the joints' driving outputs. The analysed results are important for system design and the design of the controlling mechanism for the robot.

3. Experimental tests have shown that the JL-I can implement a series of various locomotion capabilities such as 90° recovery, 180° recovery, and crossing steps. This implies the mechanical feasibility, the rationality of the analysis and the outstanding movement adaptability of the robot.

ACKNOWLEDGMENT

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